

HospiTrack Actual Deployment Steps to Render

Date: December 17, 2025

Repository: <https://github.com/JZembower/HospiTrack>

Target Platform: Render.com

Completed Pre-Deployment Steps

1. Repository Verification (Completed)

-  Checked git status: **10 commits ready to push**
-  Verified deployment files exist:
 - `render.yaml` - Render Blueprint configuration
 - `Dockerfile.prod` - Production Docker image
 - `.env.example` - Environment variable template
 - `DEPLOYMENT_GUIDE.md` - Detailed deployment guide
-  Created `.gitignore` to exclude temporary files
-  Verified data and models are in place:
 - `data/us_er.parquet` (268 KB) - Hospital dataset
 - `models/triage_model.pkl` (1.3 MB) - ML triage model
 - `models/triage_encoders.pkl` (288 bytes) - Model encoders

2. Code Preparation (Completed)

-  All bug fixes applied and tested:
 - Fixed `/api/search` lat/lon parameter passing
 - Fixed hospital name display on Explore page
 - Fixed map marker popups showing "Unknown Hospital"
 - Increased geolocation timeout from 10s to 30s
 -  Multi-page frontend fully functional:
 - Landing page (`index.html`)
 - Find Care page (`home.html`)
 - Results page with interactive map (`results.html`)
 - Explore hospitals nationwide (`explore.html`)
 - ML Triage Demo (`demo.html`)
-

Step-by-Step Deployment Guide

Step 1: Push Code to GitHub

Note: You'll need to authenticate with GitHub to push the code.

Option A: Using GitHub CLI (Recommended)

```
cd /home/ubuntu/hospitracker

# Install GitHub CLI if not already installed
# (You may already have it configured)
gh auth login

# Push the commits
git push origin main
```

Option B: Using Personal Access Token

```
cd /home/ubuntu/hospitracker

# Set up Git credentials helper (one-time setup)
git config --global credential.helper store

# Push with token authentication
# You'll be prompted for username and token
git push origin main

# When prompted:
# Username: Your GitHub username
# Password: Your Personal Access Token (not your GitHub password)
```

To create a Personal Access Token:

1. Go to GitHub.com → Settings → Developer settings → Personal access tokens → Tokens (classic)
2. Click “Generate new token (classic)”
3. Give it a descriptive name: “HospiTrack Deployment”
4. Select scopes: `repo` (full control of private repositories)
5. Click “Generate token”
6. **Copy the token immediately** (you won’t see it again)

Option C: Using SSH (If you have SSH keys set up)

```
cd /home/ubuntu/hospitracker

# Update remote to use SSH
git remote set-url origin git@github.com:JJZembower/HospiTrack.git

# Push the commits
git push origin main
```

Verify Push Success

After pushing, verify on GitHub:

```
https://github.com/JJZembower/HospiTrack/commits/main
```

You should see 10 new commits including “Add .gitignore to exclude temporary and log files”

Step 2: Deploy to Render Using Blueprint

Method 1: One-Click Deploy Button (Easiest)

1. Visit the Render Deploy URL:

<https://render.com/deploy?repo=https://github.com/JZembower/HospiTrack>

2. Connect Your GitHub Account:

- Click “Connect GitHub” if not already connected
- Authorize Render to access your repositories

3. Configure the Service:

- Render will automatically detect `render.yaml`
- **Service Name:** `hospitracker` (or choose your own)
- **Region:** Choose closest to your users (e.g., Oregon for US West)
- **Branch:** `main`
- **Plan:** Start with **Free tier** for testing

4. Review Environment Variables:

Render will auto-configure these from `render.yaml` :

- `PORT` → Auto-set by Render
- `PYTHONUNBUFFERED` → `1`
- `HOSPITRACK_DATA_PATH` → `/app/data`
- `GEOCODING_CACHE_SIZE` → `1000`
- `ML_DEMO_ENABLED` → `true`
- `LOG_LEVEL` → `INFO`

5. Click “Create Web Service”

Method 2: Manual Render Dashboard Setup (Alternative)

1. Log into Render: <https://dashboard.render.com>

2. Create New Web Service:

- Click “New +” → “Web Service”
- Connect your GitHub account
- Select repository: `JZembower/HospiTrack`
- Click “Connect”

3. Configure Build Settings:

- **Name:** `hospitracker`
- **Region:** Oregon (or preferred)
- **Branch:** `main`
- **Root Directory:** (leave blank)
- **Runtime:** Docker
- **Dockerfile Path:** `Dockerfile.prod`

4. Set Environment Variables:

Navigate to “Environment” tab and add:

`PYTHONUNBUFFERED=1`

`HOSPITRACK_DATA_PATH=/app/data`

`GEOCODING_CACHE_SIZE=1000`

`ML_DEMO_ENABLED=true`

`LOG_LEVEL=INFO`

5. Configure Health Check:

- **Path:** `/healthz`
- **Interval:** 30 seconds
- **Timeout:** 5 seconds
- **Retries:** 3

6. Deploy:

- Click "Create Web Service"
- Render will automatically start building and deploying

Step 3: Monitor Deployment

Watch Build Logs

1. Go to your Render dashboard
2. Click on your `hospitracker` service
3. View the "Logs" tab in real-time

Expected Build Process:

```
==> Building Docker image from Dockerfile.prod
==> Step 1/15 : FROM python:3.11-slim as builder
==> Step 2/15 : RUN apt-get update && apt-get install -y ...
==> Step 3/15 : COPY requirements_fastapi.txt .
==> Step 4/15 : RUN pip install --no-cache-dir -r requirements_fastapi.txt
...
==> Build successful! Starting web service...
==> Health check passed on /healthz
==> Your service is live at https://hospitracker-xxxx.onrender.com
```

Build Time: Typically 5-10 minutes for first deployment

Expected Health Check Behavior

Render will continuously check `/healthz` endpoint:

- **Status:** Should return `200 OK`
- **Response:** `{"status": "healthy", "service": "hospitracker"}`

Step 4: Get Your Public URL

Once deployment completes successfully:

1. Find Your URL:

- Go to Render dashboard → Your service
- Look for the URL at the top (e.g., `https://hospitracker-xxxx.onrender.com`)
- Click to open in a new tab

2. Save Your URL:

Public URL: `https://hospitracker-XXXX.onrender.com`
(Replace XXXX with your actual Render-generated subdomain)

Step 5: Verify Deployment

5.1 Test Landing Page

```
# Using curl
curl -I https://hospitracker-XXXX.onrender.com/

# Expected: HTTP/2 200
```

Browser Test:

1. Visit `https://hospitracker-XXXX.onrender.com/`
2. ☒ Verify landing page loads with:
 - Hero section: "What do you need most right now?"
 - Three feature cards: Find Care, Explore, Demo
 - Medical disclaimer visible

5.2 Test Find Care Page

```
curl https://hospitracker-XXXX.onrender.com/api/states
```

Browser Test:

1. Click "Find Care" or visit `/static/home.html`
2. ☒ Fill out search form:
 - Select symptom (e.g., "Chest pain")
 - Select priority (e.g., "Fastest available care")
 - Enter address: "San Francisco, CA"
 - Click "Find Hospitals"
3. ☒ Verify results page shows:
 - Interactive map with hospital markers
 - List of hospitals with details
 - Distance calculations correct

5.3 Test Explore Page

```
curl "https://hospitracker-XXXX.onrender.com/api/explore?limit=10"
```

Browser Test:

1. Visit `/static/explore.html`
2. ☒ Verify:
 - Map displays 4,088 hospitals (showing 50 at a time)
 - Hospital names display correctly (not "Unknown Hospital")
 - State filter dropdown works
 - Sorting by quality/mortality works

5.4 Test Demo (ML Triage)

```
curl -X POST https://hospitracker-XXXX.onrender.com/api/triage \
-H "Content-Type: application/json" \
-d '{
  "chief_complaint": "chest pain",
  "age": 45,
  "severity": "high",
  "heart_rate": 110,
  "systolic_bp": 160,
  "use_ml_model": true
}'
```

Browser Test:

1. Visit `/static/demo.html`
2. ☒ Fill out triage form:
 - Select complaint: "Chest pain"
 - Age: "Adult"
 - Severity: "High"
 - Enter vitals
 - Toggle ML mode ON
3. ☒ Verify response shows:
 - Triage level (e.g., "Level 2")
 - Confidence score
 - Sorting strategy recommendation
 - Medical disclaimers

5.5 Test API Documentation

```
curl https://hospitracker-XXXX.onrender.com/docs
```

Browser Test:

1. Visit `/docs` (FastAPI Swagger UI)
 2. ☒ Verify all endpoints listed:
 - GET `/healthz`
 - POST `/api/search`
 - GET `/api/explore`
 - POST `/api/triage`
 - GET `/api/states`
-

Step 6: Performance Testing

6.1 Test Response Times

```
# Health check (should be < 100ms)
time curl https://hospitracker-XXXX.onrender.com/healthz

# Hospital search (should be < 2s)
time curl -X POST https://hospitracker-XXXX.onrender.com/api/search \
  -H "Content-Type: application/json" \
  -d '{"lat": 37.7749, "lon": -122.4194, "radius": 50}'

# Explore API (should be < 1s)
time curl https://hospitracker-XXXX.onrender.com/api/explore
```

6.2 Test Under Load (Optional)

```
# Install Apache Bench if needed
sudo apt-get install apache2-utils

# 100 requests, 10 concurrent
ab -n 100 -c 10 https://hospitracker-XXXX.onrender.com/healthz
```

Step 7: Configure Custom Domain (Optional)

If you want a custom domain like `hospitracker.com` :

1. In Render Dashboard:

- Go to your service → Settings → Custom Domains
- Click “Add Custom Domain”
- Enter your domain: `hospitracker.com`

2. In Your DNS Provider:

- Add a CNAME record:

Type: CNAME

Name: @ (or www)

Value: `hospitracker-xxxx.onrender.com`

TTL: 3600

3. Wait for SSL:

- Render will automatically provision a free SSL certificate via Let’s Encrypt
- This takes 5-15 minutes

4. Verify:

bash

```
curl -I https://hospitracker.com
```

Deployment Summary

What Was Deployed

Application:

- **Name:** HospiTrack
- **Type:** FastAPI backend + Multi-page frontend
- **Runtime:** Docker (Python 3.11)
- **Database:** None (uses static Parquet file)
- **Storage:** 4.9 MB (data + models + static files)

Data Files:

- `data/us_er.parquet` (268 KB) - 4,088 US emergency rooms
- `models/triage_model.pkl` (1.3 MB) - ML model for triage prediction
- `models/triage_encoders.pkl` (288 bytes) - Label encoders

Frontend Pages:

1. **Landing Page** (/) - Multi-feature overview
2. **Find Care** (/static/home.html) - Symptom-based hospital search
3. **Results** (/static/results.html) - Interactive map with hospital list
4. **Explore** (/static/explore.html) - Browse 4,088 hospitals nationwide
5. **Demo** (/static/demo.html) - ML-powered triage simulation

API Endpoints:

- GET `/healthz` - Health check
- POST `/api/search` - Find hospitals by location and symptoms
- GET `/api/explore` - Browse hospitals with filters
- POST `/api/triage` - Get triage recommendation (rule-based or ML)
- GET `/api/states` - Get list of US states for filtering
- GET `/docs` - Interactive API documentation

Resource Usage

- **Memory:** ~512 MB (Free tier: 512 MB)
- **CPU:** Minimal (<5% average)
- **Cold Start:** ~30 seconds on free tier
- **Requests/month:** Unlimited on free tier

Troubleshooting

Issue: Build Fails - “Unable to find Dockerfile.prod”

Solution:

```
# Verify file exists in repository
git ls-files | grep Dockerfile.prod

# If missing, ensure it's committed
git add Dockerfile.prod
git commit -m "Add production Dockerfile"
git push origin main
```


Issue: Build Fails - “COPY failed: file not found”

Solution:

- Check that `data/us_er.parquet` and `models/` are in the repository
- Verify they're not excluded by `.gitignore`

```
git ls-files | grep -E "(data/|models/)"
```

Issue: Health Check Failing

Solution:

1. Check logs for application startup errors:

Render Dashboard → Logs

2. Verify `/healthz` endpoint is accessible:

bash

```
curl https://hospitracker-XXXX.onrender.com/healthz
```

3. Check environment variables are set correctly

Issue: Geocoding Timeouts

Solution:

- This is expected for some addresses due to rate limiting
- Recommend users enter lat/lon coordinates or use “Use My Location” button
- Consider upgrading to paid geocoding service (e.g., Google Maps API) for production

Issue: “No hospitals found” Despite Valid Location

Solution:

1. Check if location was geocoded successfully (view logs)
2. Increase radius slider to 100 km
3. Verify `data/us_er.parquet` contains lat/lon columns
4. Test with known coordinates:

bash

```
curl -X POST https://hospitracker-XXXX.onrender.com/api/search \
-H "Content-Type: application/json" \
-d '{"lat": 37.7749, "lon": -122.4194, "radius": 50}'
```

Issue: ML Triage Demo Returns Errors

Solution:

1. Verify `models/triage_model.pkl` exists and is not corrupted
2. Check `ML_DEMO_ENABLED=true` environment variable is set
3. View logs for model loading errors:

Render Dashboard → Logs → Search for “triage_model”

Issue: Free Tier Service Spins Down (Cold Starts)

Explanation:

- Render’s free tier spins down after 15 minutes of inactivity
- First request after spin-down takes ~30 seconds to respond

Solutions:

- Upgrade to paid plan (\$7/month) for always-on service
- Use a cron job to ping `/healthz` every 10 minutes:

```
bash
# Add to crontab
*/10 * * * * curl https://hospitracker-XXXX.onrender.com/healthz
```

Post-Deployment Recommendations

1. Monitor Performance

- Set up monitoring alerts in Render dashboard
- Configure alerts for:
 - Health check failures
 - High memory usage (>450 MB on free tier)
 - Build failures

2. Update Medical Disclaimers

- Ensure all pages display prominent medical disclaimers
- Add “Call 911 for life-threatening emergencies” warnings
- Consult legal counsel for healthcare-related software compliance

3. Consider Paid Tier for Production

Benefits of Paid Plan (\$7/month):

- No cold starts (always-on)
- Better performance (dedicated CPU)
- Custom domains with SSL
- Priority support
- More memory/CPU

4. Implement Analytics (Optional)

Add Google Analytics or Plausible to track:

- Page views per feature
- Most searched locations
- Most common symptoms
- User retention

Example (add to `<head>` of all HTML files):

```
<!-- Google Analytics -->
<script async src="https://www.googletagmanager.com/gtag/js?id=GA_TRACKING_ID"></scrip
t>
<script>
  window.dataLayer = window.dataLayer || [];
  function gtag(){dataLayer.push(arguments);}
  gtag('js', new Date());
  gtag('config', 'GA_TRACKING_ID');
</script>
```

5. Set Up CI/CD (Already Configured)

The repository includes `.github/workflows/ci.yml` for:

- Automated testing on push

- Docker build verification
- Automatic deployment to Render on main branch updates

Verify it's enabled:

<https://github.com/JZembower/Hospitrack/actions>



Quick Reference

Important URLs

Repository: <https://github.com/JZembower/Hospitrack>
 Render Dashboard: <https://dashboard.render.com>
 Public URL: <https://hospitracker-XXXX.onrender.com>
 API Docs: <https://hospitracker-XXXX.onrender.com/docs>
 Health Check: <https://hospitracker-XXXX.onrender.com/healthz>

Deployment Commands

```
# Push code
cd /home/ubuntu/hospitracker
git push origin main

# Test health
curl https://hospitracker-XXXX.onrender.com/healthz

# Test API
curl https://hospitracker-XXXX.onrender.com/api/states

# View logs (from Render dashboard or CLI)
render logs hospitracker
```

Environment Variables Reference

PORT=10000	# Auto-set by Render
PYTHONUNBUFFERED=1	# Disable Python output buffering
HOSPITRACK_DATA_PATH=/app/data	# Path to hospital data
GEOCODING_CACHE_SIZE=1000	# Number of cached locations
ML_DEMO_ENABLED=true	# Enable ML triage demo
LOG_LEVEL=INFO	# Logging verbosity



Deployment Checklist

Before marking deployment as complete, verify:

- ☐ Code pushed to GitHub successfully
- ☐ Render service created and connected to repository
- ☐ Build completed without errors
- ☐ Health check passing (`/healthz` returns 200)

- [] Landing page loads correctly
- [] Find Care search works with sample address
- [] Explore page displays all 4,088 hospitals
- [] ML Triage Demo returns predictions
- [] API documentation accessible at `/docs`
- [] No console errors in browser dev tools
- [] Mobile responsive design works on phone
- [] Medical disclaimers visible on all pages

Success!

Once all checks pass, your HospiTrack application is **live and publicly accessible!**

Share your deployment:



HospiTrack is now live!
<https://hospitracker-XXXX.onrender.com>

Find the best emergency care based on your symptoms and location.
Powered by data from 4,088 US hospitals.

Support

Issues with deployment?

- Check Render logs for detailed error messages
- Review this guide's Troubleshooting section
- Refer to `DEPLOYMENT_GUIDE.md` for additional context
- Contact Render support: <https://render.com/support>

Application bugs?

- Check `FIXES_SUMMARY.md` for known issues and fixes
- Review GitHub issues: <https://github.com/JZembower/HospiTrack/issues>
- Test locally with Docker: `docker-compose up`

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Status:  Ready for Production